Question:

Container Orchestration with Kubernetes:

- **Tool**: Kubernetes
- **Program**:
  - Set up a Kubernetes cluster (use Minikube or a cloud provider).
  - Deploy a sample application using a Deployment and Service.

Part A: Install and Start Minikube

1. Download Minikube from:
   https://minikube.sigs.k8s.io/docs/start/
2. Install Minikube and ensure Docker Desktop is running.
3. Start Minikube using Docker driver:

minikube start --driver=docker

4. Verify cluster status:
   kubectl get nodes

```
NAME       STATUS   ROLES          AGE   VERSION

minikube   Ready    control-plane   56s   v1.34.0
```

Part B: Deploy First Application (NGINX) on Minikube

1. Create an NGINX Pod:
   kubectl run nginx-pod --image=nginx

```
pod/nginx-pod created
```

2. Verify Pod:
   kubectl get pods

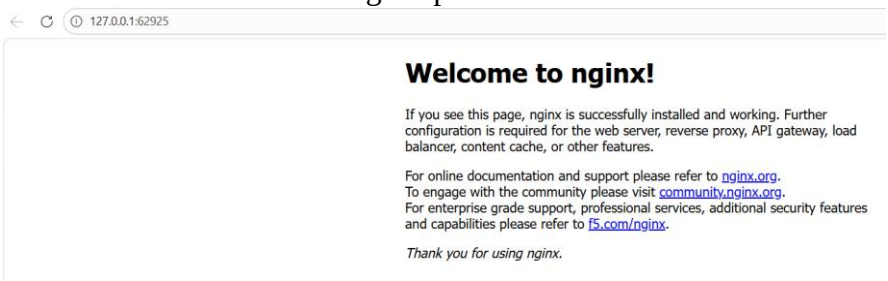
```
NAME       READY   STATUS    RESTARTS   AGE

nginx-pod   1/1     Running   0          37s
```

3. Expose Pod:
   kubectl expose pod nginx-pod --type=NodePort --port=80

```
service/nginx-pod exposed
```

4. Access application:
   minikube service nginx-pod



Part C: Deploy Your Own Flask Application on Minikube

Step 1: Create Kubernetes Deployment using Docker Hub Image

kubectl create deployment flask-deploy --image=Username/flask-docker-app

deployment.apps/flask-deploy created

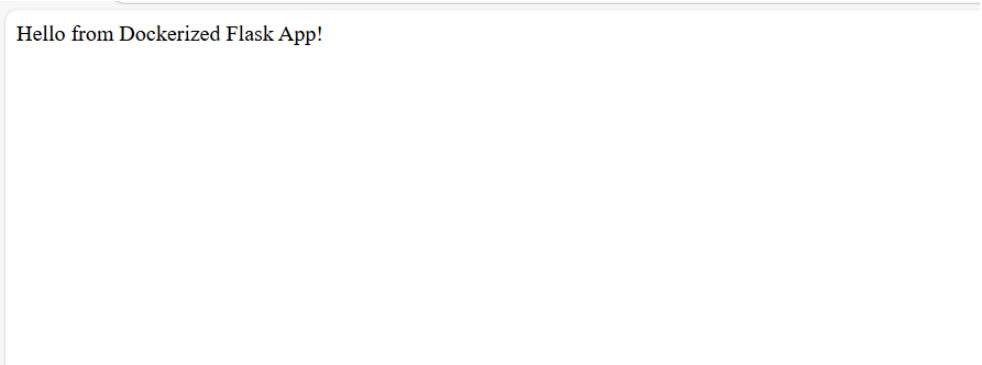
Step 2: *Expose Flask Application*

kubectl expose deployment flask-deploy --type=NodePort --port=5000

service/flask-deploy exposed

Step 3: Access Flask Application

minikube service flask-deploy

Flask application output is displayed in browser.

Test Data